DineSight (Final Year Project)

A cross-platform mobile application built with React Native and a Node.js/Firebase backend, DineSight streamlines restaurant discovery, reservations, and pre-ordering.

- **Chatbot Integration:** Developed a conversational assistant using **LangChain** to orchestrate calls to Google's Gemini API, delivering context-aware dining recommendations.
- **Recipe Feature:** Integrated TheMealDB API to fetch customized recipes based on user preferences.
- **Community Module:** Designed discussion boards for reviews, ride-pooling coordination, and group reservations.

Music Player (MERN Stack)

A full-stack web app using MongoDB, Express.js, ReactJS, and Node.js to provide a responsive music-streaming experience.

- **Spotify Integration:** Implemented OAuth-based access to Spotify's Web API for real-time song playback.
- **RESTful Services:** Built and secured endpoints for user playlists, liked songs, and playlist management.

AI-Based Disease Detection

A web application that predicts diseases from user-reported symptoms using machine-learning algorithms:

- Trained and compared Decision Trees, KNN, and Logistic Regression models for high diagnostic accuracy.
- Built a clean, scalable data pipeline and user interface to collect symptoms and display probable diagnoses.

Java Resume Builder

A desktop utility written in **Java/Swing** that allows users to input personal and professional information and export a polished PDF résumé instantly, featuring:

- Dynamic form fields with live layout preview.
- PDF generation via Java I/O without external dependencies.

Student Management Portal (ASP .NET)

An ASP .NET MVC application with **C#** and **SQL Server** for managing student records, courses, attendance, and grades:

- Role-based access control (admin, instructor, student).
- Modular service layer for easy maintenance and future feature additions.

Colormap Generation with Genetic Algorithms

A MATLAB-based digital image-processing tool that uses evolutionary algorithms to generate optimized colormaps for grayscale images:

- Defined fitness functions maximizing contrast and perceptual uniformity.
- Automated palette evolution to enhance scientific image interpretability